Binary Numbers (used to store values in Computers)

Arabic Numbering System Review

In Arabic numbering systems, like our **decimal** numbering system, there is a base value that the numbering system uses.

Base = number of unique digits

Decimal is base 10, so there are 10 digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

Also in Arabic numbering systems, the location of each digit means something.

Each position has a weight depending on the base

Value representation:
If $\mathbf{d_7 d_6 \ldots d_2 d_1 d_0}$ is a 8 digit number then:

value = $(\mathbf{d_7} \mathbf{x\ B}^7) + (\mathbf{d_6\ x\ B}^6) + \ldots + (\mathbf{d_2\ x\ B}^2) + (\mathbf{d_1\ x\ B}^1) + (\mathbf{d_0\ x\ B}^0)$

Decimal values for Powers of 10:

10^6	10^5	10^4	10^3	10^2	10^1	10^0
1000000	**100000**	**10000**	**1000**	**100**	**10**	**1**

Decimal (Base 10) value example:

Decimal number **3254** can be broken out as follows:

Thousands	Hundreds	Tens	Units
3	2	5	4

Using exponential representation we have the value:

$$\begin{aligned} 3254 &= (3 \times 10^3) + (2 \times 10^2) + (5 \times 10^1) + (4 \times 10^0) \\ &= (3 \times 1000) + (2 \times 100) + (5 \times 10) + (4 \times 1) \end{aligned}$$

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Binary numbers (base-2)

In the binary system, the base **2** is substituted for the base **10**.

Decimal values for Powers of 2:

2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
256	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**

Note: You can easily extend this table to the **left** by continuing to multiply by 2.

Example:
The binary number **1110** can be analyzed and converted to the decimal number 14 as follows:

$$1110_2 = (1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0)$$
$$= (1 \times 8) + (1 \times 4) + (1 \times 2) + (0 \times 1) \quad = 14_{10}$$

Note the use of the subscripts 2 and 10, indicating the base, to eliminate any confusion in a term or expression.